

HW:- 15/07/26

1) Find the frequency counts for the following codes-

(i) for ($i=1; i \leq n; i++$);
printf("Hello");

(ii) $i=1;$
 $i=i+20;$
printf("%d", i);

(iii) $i=1;$
while ($i \leq n$)
{
printf("%d", i);
 $i++$;
}

2) For all the shell programs done in OS lab, find out time complexity using frequency count.

1) (i)

code	cost	repetition	Total
$i=1$	1	1	1
$i \leq n$	1	n	n
$i++$	1	n	n
printf("Hello");	1	1	1
<u><u>$2n+2$</u></u>			

(ii)

code	cost	Repetition	Total
$i=1;$	1	1	1
$i=i+20;$	2	1	2
printf("%d", i);	1	1	1
<u><u>4</u></u>			

(iii)

code	cost	Repetition	Total
$i=1;$	1	1	1
while ($i \leq n$) {	1	$n+1$	$n+1$
printf("%d", i);	1	n	n
$i++$; }	1	n	n
<u><u>$3n+2$</u></u>			

2.) Basic shell program $\Rightarrow$

```
#!/bin/bash
```

```
echo -n "Enter the first number:"
read num1
```

```
echo -n "Enter 2nd number:"
read num2
```

```
sum=$((num1+num2))
```

~~echo "the sum of \$num1 and \$num2 is: \$sum"~~

echo "the sum of \$num1 and \$num2 is: \$sum"

code	cost	Repetition	Total
echo -n "Enter the first number: "	1	1	1
read num1	1	1	2
echo -n "Enter 2nd number: "	1	1	3
read num2	1	1	4
sum=\$((num1+num2)) (assignment + 1 addition)	2	1	6
echo "the sum of \$num1 and \$num2 is: \$sum"	1	1	7